INTERNATIONAL SEARCH REPORT

International application No.
PCT/US2022/030487

A. CLASSIFICATION OF SUBJECT MATTER

G01N 33/487(2006.01)i

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01N 33/487(2006.01); B01D 29/50(2006.01); B64D 15/06(2006.01); B81B 1/00(2006.01); C12Q 1/68(2006.01); C21C 7/00(2006.01); G01N 27/447(2006.01); H01F 1/44(2006.01); H01M 8/0206(2016.01); H01M 8/0215(2016.01)

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Korean utility models and applications for utility models
Japanese utility models and applications for utility models

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

eKOMPASS(KIPO internal) & Keywords: controlling particle movement in fluid, field-responsive viscosity, speed controller, electro-rheological fluid, magneto-rheological fluid, solenoid

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 2013-0344498 A1 (BOREAL GENOMICS CORP.) 26 December 2013 (2013-12-26) See paragraphs [0002], [0020], [0078], [0167], [0191]-[0192], [0282]-[0291], [0308], [0380], [0441]; and figures 10A-10D.	1-3,8-14,18,21-23
Y		4-7,15-17,19-20
Y	US 2019-0096555 A1 (MASSACHUSETTS INSTITUTE OF TECHNOLOGY) 28 March 2019 (2019-03-28) See abstract; paragraphs [0065]-[0078]; and claims 1, 4.	4-7,15-17,19-20
A	US 2006-0038328 A1 (LU, JIAN et al.) 23 February 2006 (2006-02-23) See the whole document.	1-23
A	KR 10-2012-0117834 A (CYTOVERA INC.) 24 October 2012 (2012-10-24) See the whole document.	1-23

☑ Further documents are listed in the continuation of Box C. ☑ See patent family annex.

* Special categories of cited documents:
"A" document defining the general state of the art which is not considered to be of particular relevance
"D" document cited by the applicant in the international application
"E" earlier application or patent but published on or after the international filing date
"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
"O" document referring to an oral disclosure, use, exhibition or other means
"P" document published prior to the international filing date but later than the priority date claimed
"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
04 November 2022	**07 November 2022**

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C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	KR 10-2020-0027034 A (24M TECHNOLOGIES, INC.) 11 March 2020 (2020-03-11) See the whole document.	1-23

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/US2022/030487

Patent document cited in search report	Publication date (day/month/year)	Patent family member(s)	Publication date (day/month/year)
US 2013-0344498 A1	26 December 2013	EP 2458004 A1	30 May 2012
		EP 2458004 B1	15 March 2017
		JP 2007-526823 A	20 September 2007
		JP 5485219 B2	07 May 2014
		US 10975421 B2	13 April 2021
		US 2009-0139867 A1	04 June 2009
		US 2021-0238663 A1	05 August 2021
		US 8133371 B2	13 March 2012
		WO 2005-072854 A1	11 August 2005
US 2019-0096555 A1	28 March 2019	US 11079141 B2	03 August 2021
		US 2015-0179321 A1	25 June 2015
		WO 2015-095660 A1	25 June 2015
US 2006-0038328 A1	23 February 2006	EP 1294510 A1	26 March 2003
		EP 1294510 B1	13 July 2005
		JP 2003-534920 A	25 November 2003
		US 2002-0186616 A1	12 December 2002
		US 6402367 B1	11 June 2002
		WO 01-91949 A1	06 December 2001
KR 10-2012-0117834 A	24 October 2012	BR 112012018376 A2	10 October 2017
		CN 102791616 A	21 November 2012
		CN 102791616 B	29 July 2015
		EP 2516320 A1	31 October 2012
		JP 2013-515599 A	09 May 2013
		JP 5624629 B2	12 November 2014
		KR 10-1443133 B1	03 November 2014
		US 2012-0258459 A1	11 October 2012
		US 8679751 B2	25 March 2014
		WO 2011-079217 A1	30 June 2011
KR 10-2020-0027034 A	11 March 2020	CN 102119461 A	06 July 2011
		CN 102119461 B	03 August 2016
		EP 2297810 A1	23 March 2011
		EP 2297810 B1	24 December 2014
		JP 2011-524074 A	25 August 2011
		JP 5575757 B2	20 August 2014
		KR 10-1721269 B1	29 March 2017
		KR 10-2011-0056373 A	27 May 2011
		KR 10-2022-0137172 A	11 October 2022
		KR 10-2281072 B1	23 July 2021
		US 10236518 B2	19 March 2019
		US 2010-0047671 A1	25 February 2010
		US 2019-0355998 A1	21 November 2019
		US 8722226 B2	13 May 2014
		WO 2009-151639 A1	17 December 2009